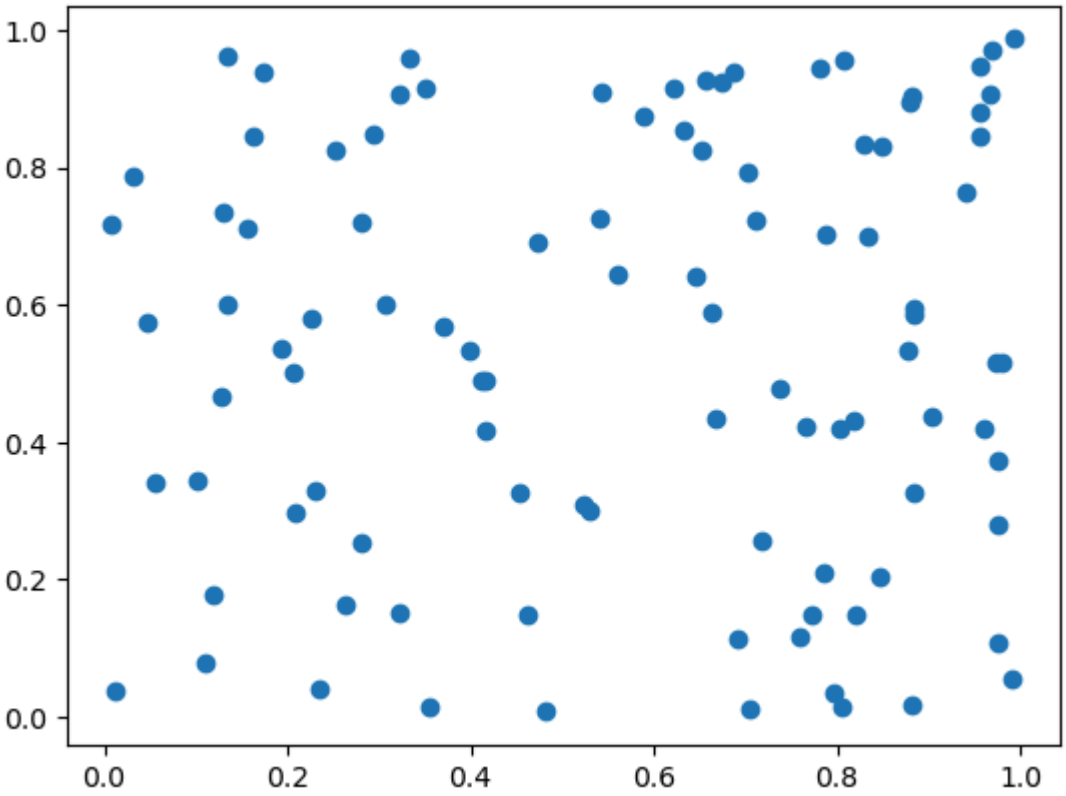
In [5]:
```
import numpy as np
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt
```

In [6]:
```
X=np.random.rand(100,2)
```

In [7]:
```
plt.scatter(X[:,0], X[:,1])
plt.show()
```



In [9]:
```
kmeans=KMeans(n_clusters=3, random_state=0)
kmeans.fit(X)
```

```
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.
  warnings.warn(
```

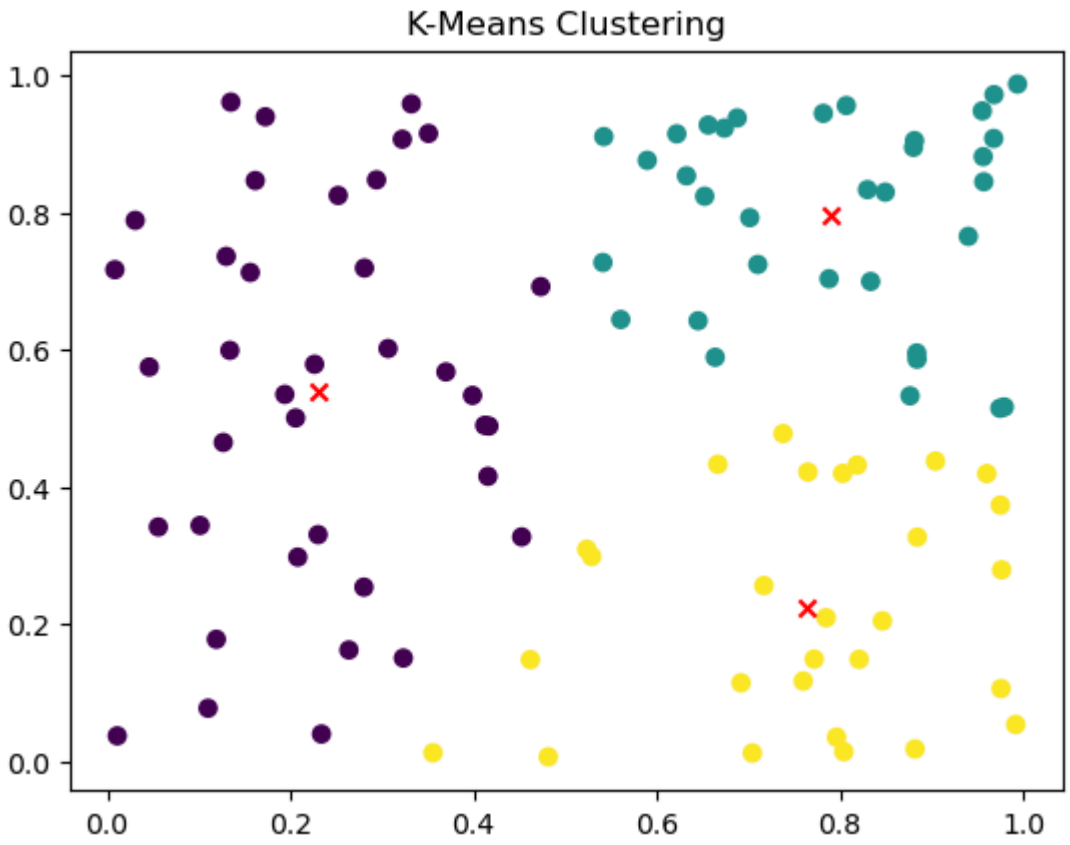
Out[9]:

▾ KMeans

▶ Parameters

In [10]:
```
centroids=kmeans.cluster_centers_
labels = kmeans.labels_
```

In [11]:
```
plt.scatter(X[:, 0], X[:,1], c=labels)
plt.scatter(centroids[:, 0], centroids[:, 1], marker='x', color='red')
plt.title("K-Means Clustering")
plt.show()
```



In [ ]: